Real-Time Ingestion Architecture Guide

1. Overview of Ingestion System

SentinelGrid operates on a schema-based log analyzer that parses events via unsupervised machine learning (Isolation Forest). To transition from simulation logs to live environment streams, production endpoints must forward logs conforming to the structured SentinelGrid schema.

2. Log Event Schema Format

Field Name	Data Type	Description & Examples
timestamp	String (ISO8601)	ISO timestamp. e.g., '2026-07-02T16:18:21+05:30'
username	String	User identifier or system account name. e.g., 'dr_sharma'
event_type	String	Event class: 'login', 'access_server', 'privilege_change', 'data_transfer'
ip_address	String (IPv4)	Origin IP of request. e.g., '95.213.255.1'
country	String	Source country geolocated from IP. e.g., 'Russia'
city	String	Source city context. e.g., 'Moscow'
server	String	Target system asset name. e.g., 'backup_server'
data_transferred_mb	Float	Payload transfer size in megabytes. e.g., 4500.0
privilege_level	String	Account clearance role: 'user' or 'admin'
status	String	Outcome of operation: 'success' or 'failed'

3. API Ingestion Endpoint (Live Integration)

Logs should be POSTed in JSON format to the endpoint: http://localhost:8000/api/logs/ingest.
Your streaming agent (e.g. FluentBit, Logstash) should send a JSON payload structured matching the schema fields above. The API immediately feeds this to the model.

4. Stream Collector Configurations

- Windows Systems: Use Windows Event Forwarding (WEF) combined with NXLog to capture Security Event IDs (4624 for Login, 4720 for User Creation) and convert to SentinelGrid format.
- Linux Servers: Configure Rsyslog to parse sshd logs and auditd logs, forwarding as JSON packages.
- Network Devices: Forward syslog messages to a centralized Logstash container for mapping.